DAD Test 2 Sample Practice Solutions-1

Business Digitalisation (Swinburne University of Technology (Vietnam))



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INF10025 Data Management and Analytics

Test 2 – Sample Practice Questions – Suggested Answers

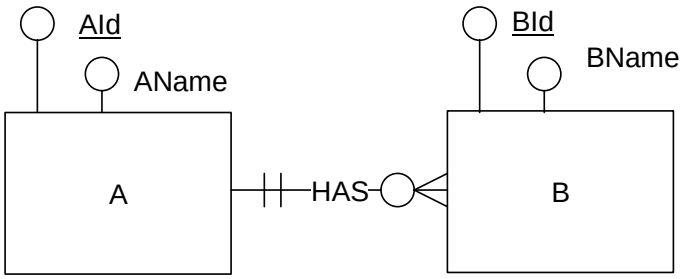
The test questions will be a combination of

- Fill in the blank
- Multiple Choice
- Combine the code segments (see Q21)

The questions cover all weeks 1-7 but primarily concentrating on weeks 5 to 7 (but no SQL aggregate functions / group by / having).

These questions are typical of those that may be found in the test.

1. Consider the following ER Diagram:



a) What is the identifier of the Entity named A?

Answer: _AID_____

b) What is the identifier of the Entity named B?

Answer: __BID_____

c) What is the FK (if any) in any of the following Entities when converted to a relational schema?

Entity A: _____

Entity B: __AID references A_____

d) Write the relational schema based on this ERD. Indicate all PKs and FKs

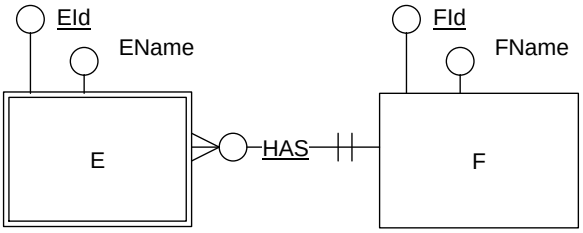
A(Aid, AName)
PK (Aid)

B(Bid, BName, Aid)
PK (Bid)
FK (Aid) references B

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2. Consider the following ER Diagram:



a) What is the identifier of the Entity named E?

Answer: _Fid + Eid_____

b) What is the identifier of the Entity named F?

Answer: Fid_____

c) What is the FK (if any) of the following Entities when converted to a relational schema?

Entity E: _Fid_____

Entity F: _____

d) Write the relational schema based on this ERD. Indicate all PKs and FKs

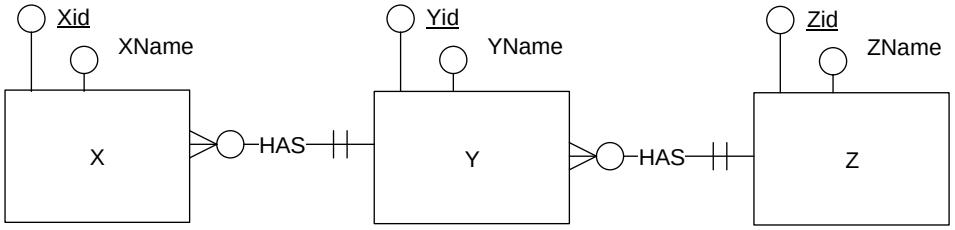
F(Fid, FName)
PK (Fid)

E(Fid, Eid, EName)
PK (Fid, Eid)
FK (Fid) references F

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3. Consider the following ER Diagram:



a) What is the identifier of the Entity named X?

Answer: _XID_____

b) What is the identifier of the Entity named Y?

Answer: _____YID_____

c) What is the identifier of the Entity named Z?

Answer: __ZID_____

d) What is the FK (if any) of the following Entities when converted to a relational schema?

Entity X: __YID_____

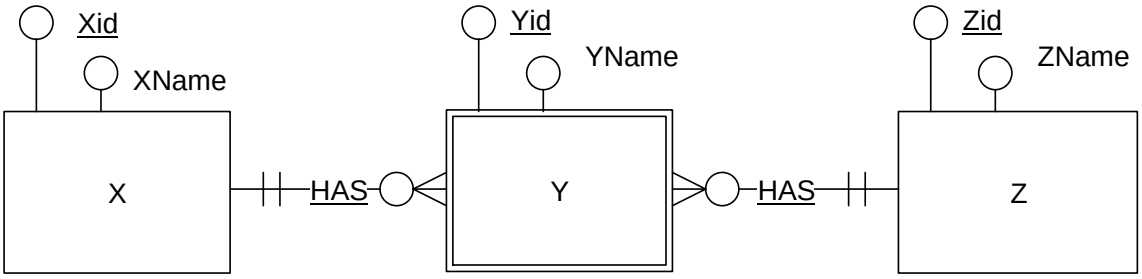
Entity Y: ___ZID_____

Entity Z: _____

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4. Consider the following ER Diagram:.



a) What is the identifier of the Entity named X?

Answer: ____XID_____

b) What is the identifier of the Entity named Y?

Answer: _____YID, XID, ZID_____

c) What is the identifier of the Entity named Z?

Answer: ____ZID_____

d) What is the FK (if any) of the following Entities when converted to a relational schema?

Entity X: _____

Entity Y: ____FK1: XID FK2: ZID_____

Entity Z: _____

e) Write the relational schema based on this ERD. Indicate all PKs and FKs

X(Xid, Xname)
PK (Xid)

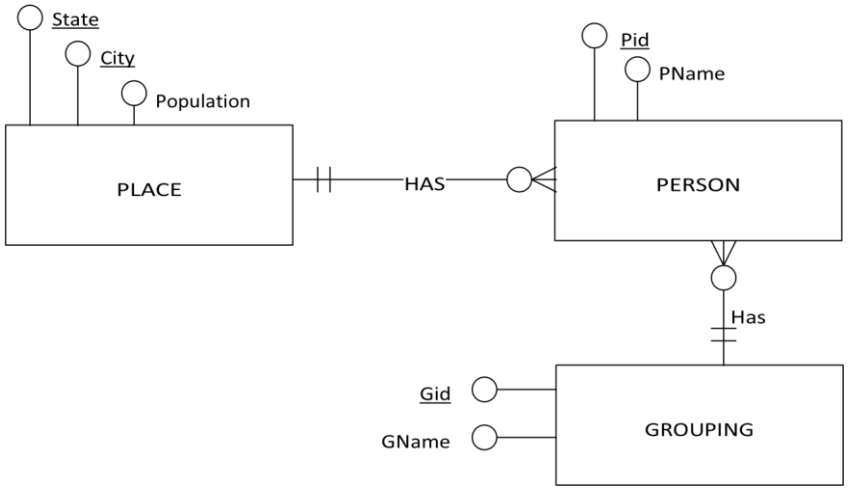
Z(Zid, Zname)
PK(Zid)

Y(Xid, Zid, Yid, YName)
PK(Xid, Zid, Yid)
FK(Xid) references X
FK(Zid) references Z

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5. Consider the following ER Diagram:.



When converted to a relational schema

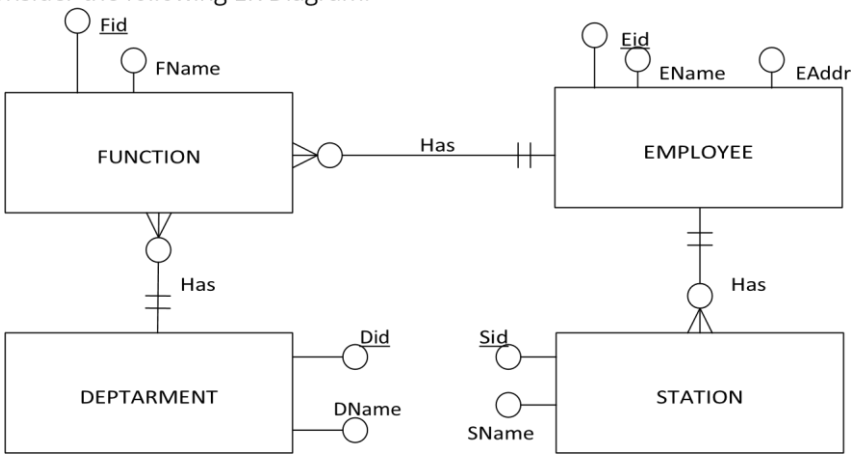
a. How many Foreign Keys are there in total?

Answer (circle one) 0 1 **2** 3 4 5

b. Which relation has the most Foreign Keys?

Answer (circle one) Place **Person** Grouping

6. Consider the following ER Diagram:



When converted to a relational schema

a. How many tables will have no Foreign Keys?

Answer (circle one) 0 1 **2** 3 4

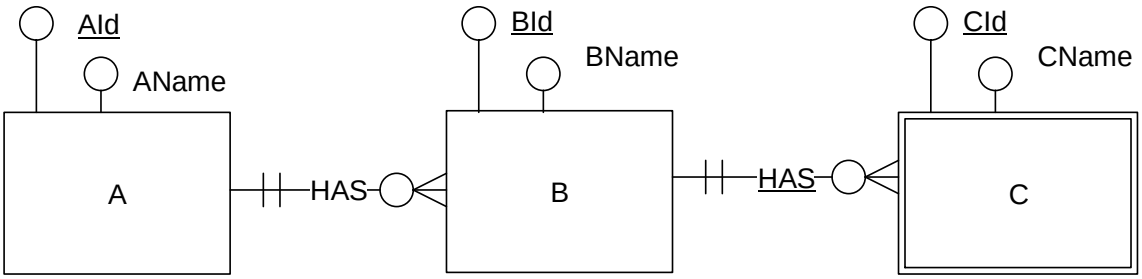
b. Which table will have the most attributes?

Answer (circle one) Department **Function** Employee Station

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7. Consider the following ER Diagram:.



a. What is the identifier of the Entity named A?

Answer: ___Aid_____

b. What is the identifier of the Entity named B?

Answer: ____Bid_____

c. What is the identifier of the Entity named C?

Answer: __Bid Cid_____

d. What is the FK (if any) of the following Entities when converted to a relational schema?

Entity A: _____

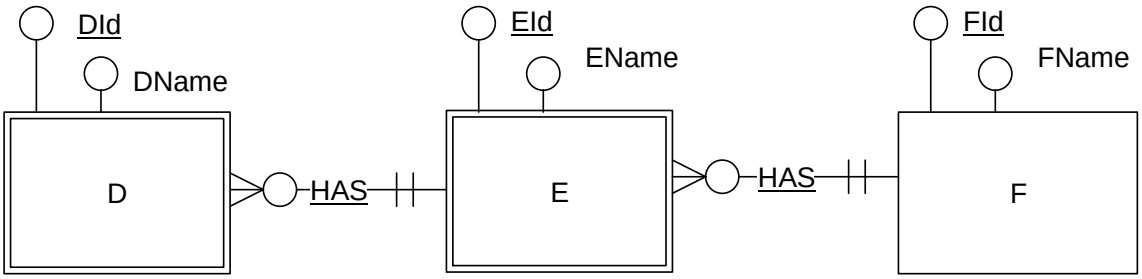
Entity B: __Aid_____

Entity C: ___Bid_____

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8. Consider the following ER Diagram:.



a) What is the identifier of the Entity named D?

Answer: ___Fid Eid Did_____

b) What is the identifier of the Entity named E?

Answer: _____Fid Eid_____

c) What is the identifier of the Entity named F?

Answer: _____Fid_____

d) What is the FK (if any) of the following Entities when converted to a relational schema?

Entity D: _____Fid Eid references E_____

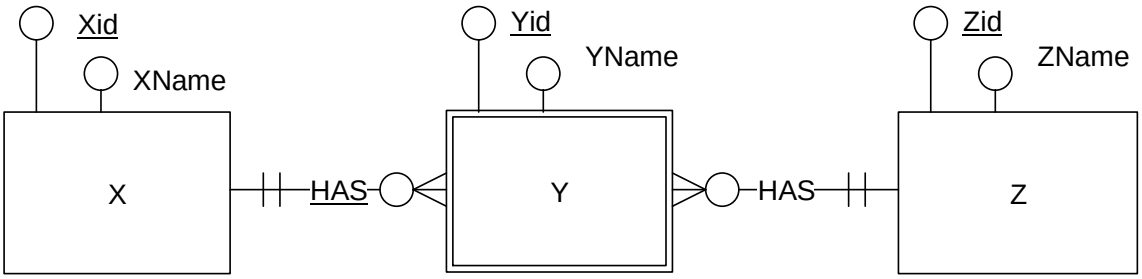
Entity E: _____Fid references F_____

Entity F: _____

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9. Consider the following ER Diagram:.



a) What is the identifier of the Entity named X?

Answer: _____Xid_____

b) What is the identifier of the Entity named Y?

Answer: _____Xid Yid_____

Note that the relationship between Y & Z is NOT underlined.

c) What is the identifier of the Entity named Z?

Answer: _____Zid_____

d) What is the FK (if any) of the following Entities when converted to a relational schema?

Entity X: _____

Entity Y: _____Xid references X Zid references Z_____

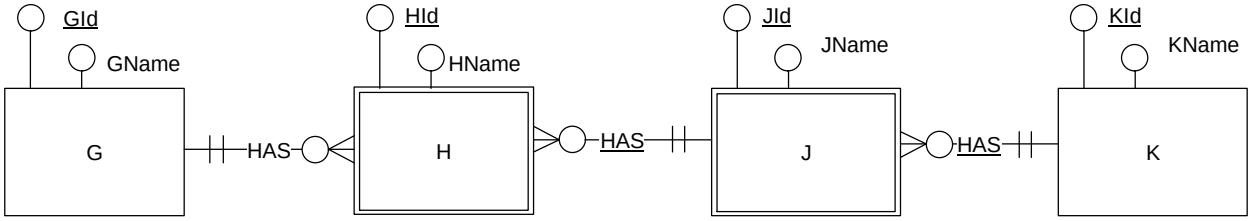
Entity Z: _____

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10.

Consider the following ER Diagram:



a) What is the identifier of the Entity named K?

Answer: _Kid_____

b) What is the identifier of the Entity named J?

Answer: ____ Kid_Jid_____

c) What is the identifier of the Entity named H?

Answer: ____Kid Jid Hid_____

d) What is the identifier of the Entity named G?

Answer: _____Gid_____

e) What is the FK (if any) of the following Entities when converted to a relational schema?

Entity G: _____

Entity H: _____Kid , Jid references J and Gid References G_____

Entity J: _____Kid references K_____

Entity K: _____

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11.

Consider the following tables and data:

GRADE (gradeID, description)

Primary Key (gradeID)

TYPE (TypeCode, title, gradeID, catid)

Primary Key (TypeCode)

Foreign Key (gradeID) references GRADE

CUSTOMER (custno, custname, typecode, sales, location)

Primary Key (custno)

Foreign Key (typecode) references TYPE

CUSTNO	CUSTNAME	TYPECODE	SALES	LOCATION
21	Wendy	1	300	Haw
24	Sue	3	700	Haw
27	Nick	2		Kew
22	Dave	4	95	Richmond
25	Jenny	1	525	Haw
29	Bob	4	15	Kew
23	Helen	3	200	Haw
28	Linda	1	25	Kew
26	Xena	3		Richmond

TYPECODE	TITLE	GRADEID	CATEGORY
1	Under 20	WC	X
2	Under 20	M	Z
3	20 plus	WC	Z
4	20 plus	M	X
5	Special	M	S

GRADEID	DESCRIPTION
WC	Women and Children
M	Men

** - see back page for SQL DDL & DML statements*

ANSWERS TO QUESTION 11-13 can be found by executing the SQL statements on the final page of this test and then running each of the SELECT statements in your Oracle account.

11. How many rows are returned by the following SQL statements?

a. Select * from customer where location = 'Kew'; **3**

b. Select * from customer where location = 'HAW'; **0**

c. Select * from customer where location = 'Haw'; **4**

d. Select * from customer where Upper(location) = 'Haw'; **0**

e. Select * from customer where Lower(location) = 'haw'; **4**

f. Select * from customer where location = 'Kew' AND sales < 100; **2**

g. Select * from customer where location = 'KEW'; **0**

h. Select * from customer where location = 'haw'; **0**

i. Select * from customer where Upper(location) = '%H%'; **6**

j. Select * from customer where location = 'Kew' OR sales < 200; 4

k. Select * from customer where sales < 100 OR sales > 400 AND typecode > 1; 4

l. Select * from customer where (sales < 100 OR sales > 400) AND typecode > 1; 3

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m. Select * from customer where typecode = 1 OR typecode = 3; 6

n. Select * from customer where typecode = 1 OR typecode = 3 AND sales > 500 4

o. Select * from customer where sales >= 0 OR sales <= 500 0 AND location = 'Haw' 7

p.
```
Select * from Customer
 where UPPER(custname) like '%d%' or
LOWER(custname) like '%N%'
```
0

q.
```
Select c.custname, t.title
 from customer c
 inner join type t
 on c.typecode = t.typecode;
```
9

r.
```
Select c.custname, t.title
from customer c
 Inner join type t

 on c.typecode = t.typecode
where sales < 100 ;
```
3

s.
```
Select c.custname, t.title
 from customer c
 inner join type t

 on c.typecode = t.typecode

where sales > 100 OR sales < 400   and t.typecode = 1;
```
1

t.
```
Select c.custname, t.title, g.description
from customer c
 inner join type t
on c.typecode = t.typecode
inner join grade g
 on  t.gradeid = g.gradeid;
```
9

12. Which name is listed 2nd when this SQL statement is executed? Bob Dave Nick Sue Xena

```
SELECT custname
FROM CUSTOMER
WHERE typecode >1
ORDER BY custname
```

13. Which Location is listed last when this SQL statement is executed? 1 Haw 3 Haw

```
SELECT *                                  1Kew    3  Rich
FROM CUSTOMER                             2 Kew   4 Kew
ORDER BY typecode, location               3Haw    4 Rich
```

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14. Write a single SQL statement to list all customer names and sales amounts in **descending** sales amount sequence

```
SELECT custname, sales
FROM  customer
ORDER BY sales DESC
```

15. Write a single SQL statement to list all customer numbers, customer names and locations in **ascending location sequence**. If two or more rows have the same location, list those rows in ascending customer name sequence

Haw	23	Helen
Haw	25	Jenny
Haw	24	Sue
Haw	21	Wendy
Kew	29	Bob
Kew	28	Linda
Kew	27	Nick
Richmond	22	Dave
Richmond	26	Xena

```
SELECT location, custno, custname
FROM  customer
ORDER BY location ASC, custname ASC
```

16. Write a **single** SQL statement to list the names of customers who have a sales value in one the following ranges:

a. 25-100 inclusive

b. 500-900 inclusive

```
SELECT custname
FROM  customer
WHERE (sales >= 25 AND sales <=100)
OR (sales >=500 AND sales <=900)
```

17. Write a single SQL statement to list the names of customers have the letter N in their name. Allow for both upper case and lower case characters.

```
SELECT custname
FROM  customer
WHERE UPPER(custname) LIKE '%N%'
```

18. Write a single SQL statement to list the customer name of any customer that has a null sales amount value.

```
SELECT custname
FROM  customer
WHERE sales IS NULL
```

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19. Write a single SQL statement to list each customer name and the title that he/she has

CUSTNAME	TYPENAME
Wendy	Under 20
Sue	20 plus
Nick	Under 20
Dave	20 plus
Jenny	Under 20
Bob	20 plus
Helen	20 plus
Linda	Under 20
Xena	20 plus

```
SELECT c.custname , t.title AS "TYPENAME"
FROM  customer c,
INNER JOIN type t
 ON c.typecode = t.typecode
```

20. Write a single SQL statement to list each customer name only if they have a title that contains the word 'Under'

CUSTNAME	TYPECODE	TYPENAME
Wendy	1	Under 20
Nick	2	Under 20
Jenny	1	Under 20
Linda	1	Under 20

```
SELECT c.custname , t.typecode, t.title AS "TYPENAME"
FROM  customer c,
INNER JOIN type t
 ON c.typecode = t.typecode
WHERE UPPER(t.title) LIKE '%UNDER%'
```

ANSWERS TO QUESTION 21-22 can be found by executing the SQL statements on the final page of this test and then running each of the SELECT statements in your Oracle account.

21. Complete this SQL statement from the parts below that lists the customer name and type title for every row in the customer table.

```
SELECT C.CUSTNAME, T.TITLE
FROM  CUSTOMER C
INNER JOIN TYPE T
```

a. ON C.CUSTNAME = T.TITLE

b. ON T.CUSTNAME = T.TITLE

c. ON T.CUSTNAME = C.TITLE

d. ON C.TYPECODE = T.TYPECODE

e. ON T.TYPECODE = T.TYPECODE

f. ON C.TYPECODE = C.TYPECODE

g. ON C.CUSTNO = T.TYPECODE

h. ON C.CUSTNO = C.TYPECODE

i. ON T.CUSTNO = T.TYPECODE

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22. How many rows will be selected by this SQL statement?

```
SELECT T.TITLE, G.DESCRIPTION
FROM TYPE T
INNER JOIN GRADE G
ON T.GRADEID = S.GRADEID
```

ANSWER **0 1 2 3 4 5 6 7 8 9 10**

You may use the SQL code below to create the tables and populate them with data so that you can check your answers to the above questions.

SQL DDL statements and SQL DML statements

```
Drop table customer;
Drop table type;
Drop table grade;

create table GRADE ( gradeID
varchar(2), description varchar(20),
Primary Key (gradeID) );

create table TYPE ( TypeCode
number,
Title      varchar2(20),
gradeID varchar(2), category
varchar(1),  Primary Key
(TypeCode),
Foreign Key (gradeID) references GRADE );

create  table  customer(
custno            number,
custname    varchar(20),
typecode number,  sales
number(7,2),      location
varchar(20),  Primary Key
(custno),
 Foreign Key (typecode) references TYPE );

Insert into grade (gradeID,description) values ('WC','Women and Children');
Insert into grade (gradeID,description) values ('M','Men');
Insert into TYPE (TypeCode,Title,gradeID,category) values (1,'Under 20','WC','X');
Insert into TYPE (TypeCode,Title,gradeID,category) values (2,'Under 20','M','Z');
Insert into TYPE (TypeCode,Title,gradeID,category) values (3,'20 plus','WC','Z');
Insert into TYPE (TypeCode,Title,gradeID,category) values (4,'20 plus','M','X');
Insert into TYPE (TypeCode,Title,gradeID,category) values (5,'Special','M','S');
Insert into customer (custno, custname, TypeCode,sales,location) values (21,'Wendy',1,300,'Haw');
Insert into customer (custno, custname, TypeCode,sales,location) values (24,'Sue',3,700,'Haw');
Insert into customer (custno, custname, TypeCode,sales,location) values (27,'Nick',2,NULL,'Kew');
Insert into customer (custno, custname, TypeCode,sales,location) values (22,'Dave',4,95,'Richmond');
Insert into customer (custno, custname, TypeCode,sales,location) values (25,'Jenny',1,525,'Haw');
Insert into customer (custno, custname, TypeCode,sales,location) values (29,'Bob',4,15,'Kew');
Insert into customer (custno, custname, TypeCode,sales,location) values (23,'Helen',3,200,'Haw');
Insert into customer (custno, custname, TypeCode,sales,location) values (28,'Linda',1,25,'Kew');
Insert into customer (custno, custname, TypeCode,sales,location) values (26,'Xena',3,Null, 'Rich
```

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